Self-Check

3. What role does Oracle VirtualBox play in the penetration-testing lab you are setting up?
   a. Master operating system
   b. Popular cyber attack tool
   c. Host to the lab targets and Kali Linux attack machines
   d. Download facilitator
4. A snapshot captures the current state and configuration of a virtual machine.
   a. True
   b. False

□ Check your answers at the end of this chapter.

Setting Up Targets

After setting up the Kali Linux VM as the attack machine, you are ready to set up the targets for pen testing. In this section, you create seven targets, one each for Metasploitable, Windows 7, Windows 10, Windows 11, Windows Server 2022, DVWA, and the Axigen mail server.

Exam Tip | This section of the chapter prepares targets for your penetrating testing activities. Later in this course, you gather information, scan for vulnerabilities, perform attacks, and exploit vulnerabilities. These activities are detailed in the CompTIA PenTest+ exam objectives domains 2.0 and 3.0. You need a variety of targets in your penetrating testing lab to perform these activities.

Creating a Metasploitable Target

Metasploitable2 has been purposefully constructed to be vulnerable to attack. It is designed to provide pen testers with a target containing security flaws for practicing pen testing. You download Metasploitable2, provided by Rapid7, in the following steps. Rapid7 also provides an exploitability guide that outlines Metasploitable2's security flaws (https://docs.rapid7.com/metasploit/metasploitable-2-exploitability-guide).

Note 3 | You can find a video showing how to install Metasploitable in VirtualBox at https://www.youtube.com/watch?v=1rIvnMenA2g. If the URL no longer works, you can perform a search on YouTube for "installing metasploitable 2 on virtualbox" to find another video.

Although Metasploitable is not an OVA, Rapid7 provides all the necessary files to build a new VM in VirtualBox.

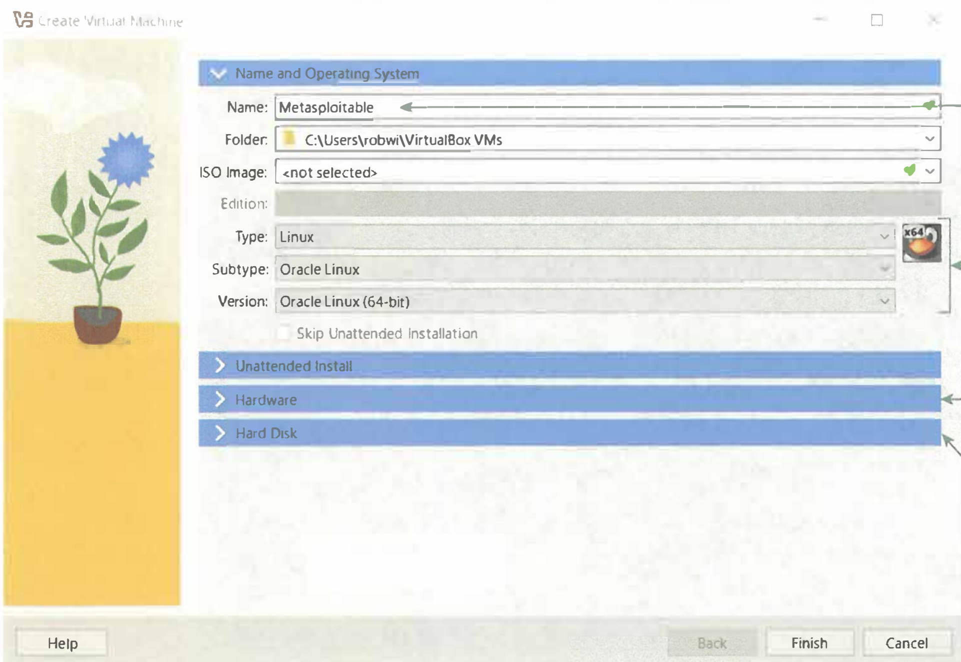
To create the Metasploitable2 VirtualBox VM:

1. Use a browser to go to **https://www.rapid7.com/products/metasploit/metasploitable/** and then click the **DOWNLOAD NOW** button to download Metasploitable2 as a compressed zip file named **metasploitable-linux-2.0.0.zip**.
2. In File Explorer, right-click the downloaded file, and then click **Extract All** to extract the installation files contained in the zip file. If you want to keep all your downloads in a different folder (such as C:\PenTestDownloads), move the zip file there first, and then perform the extraction.
3. Start VirtualBox with administrative permissions. If you are logged in with an administrative account, then simply start VirtualBox. If you aren't logged in with an administrative account, then right-click VirtualBox and choose "Run as administrator". If you cannot perform certain VirtualBox

operations because buttons or features are not active, it's probably because you failed to start VirtualBox with administrative permissions.

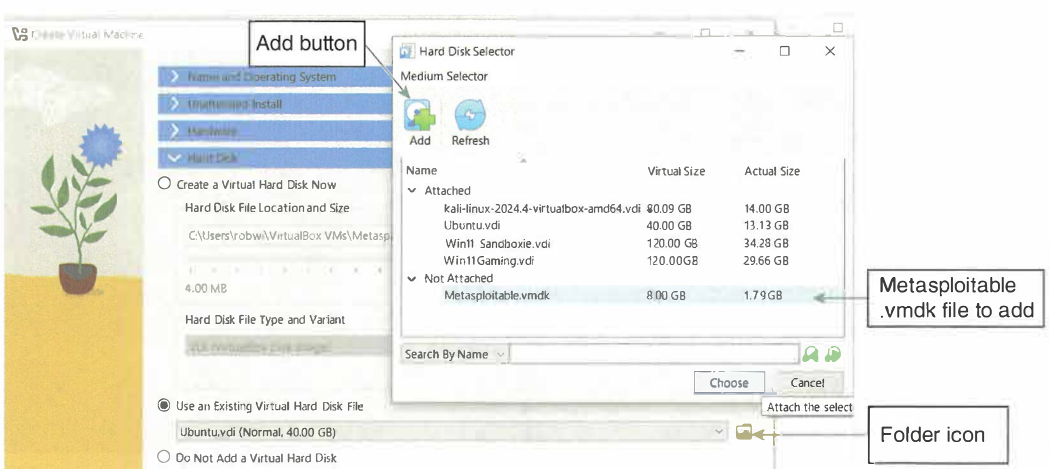
4. In VirtualBox, click **Machine** on the menu bar, and then click **New**. In the Create Virtual Machine dialog box, in the Name and Operating System section, enter **Metasploitable** as the Name. Select **Linux** as the Type, and **Oracle Linux (64-bit)** as the Version (Figure 2-8).

Figure 2-8 Entering the name and operating system of the Metasploitable target



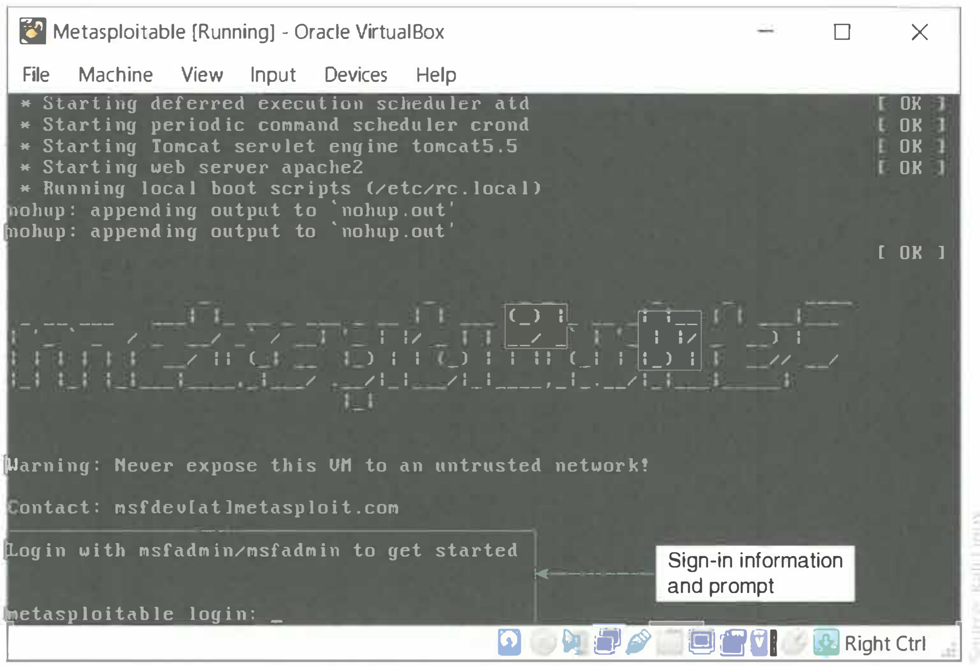
5. Click the **Hardware** button to expand the Hardware configuration section and enter **512** MB for the Base Memory size of the VM. This VM doesn't require the default 2 GB.
6. Click the **Hard Disk** button and then select the **Use an existing virtual hard disk file** option button if it is not already selected.
7. Click the folder icon in the Hard Disk section to open the Hard Disk Selector dialog box.
8. Click the **Add** button, navigate to and select the **Metasploitable.vmdk** file in the extracted folder, and then click **Open** to add this file to the list of hard disks to use it with the Metasploitable2 VM.
9. In the Metasploitable – Hard Disk Selector dialog box, select the **Metasploitable.vmdk** file in the list of hard drives (Figure 2-9), and then click **Choose**.

Figure 2-9 Selecting the Metasploitable.vmdk file



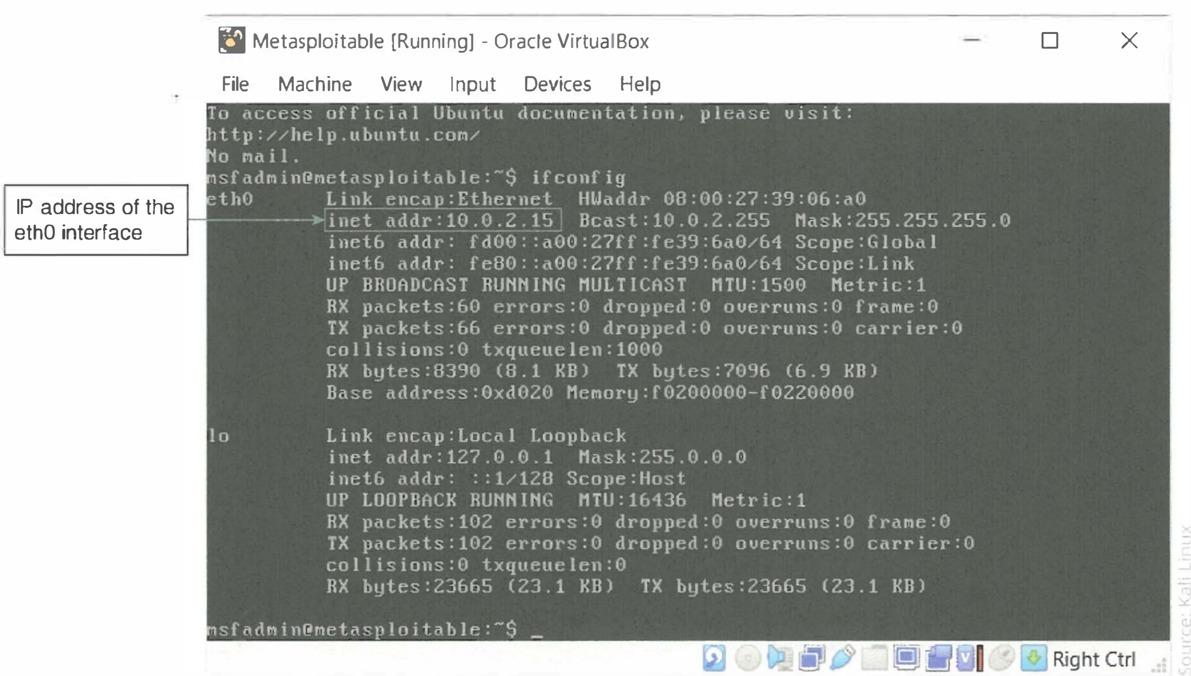
10. In the Create Virtual Machine dialog box, click the **Finish** button to create the VM as an administrator.
11. In VirtualBox, right-click the Metasploitable VM in the left pane, click **Start** on the shortcut menu, and then click **Normal Start** to start the Metasploitable VM and display the login information (Figure 2-10).

Figure 2-10 Metasploitable login screen with login info



12. Sign in to the Metasploitable VM using **msfadmin** as the username and **msfadmin** as the password.
13. Type **ifconfig** and press **Enter**. Note the IP address of the eth0 interface (Figure 2-11). You use this address to target the VM for pen testing.

Figure 2-11 Using ifconfig to determine the Metasploitable2 VM IP address



14. Use the VirtualBox Manager window to change the network adapter settings for the Metasploitable VM to a host-only adapter so that this VM is on the same network as your other VMs.
15. Take a snapshot of the Metasploitable VM so that you can reset to the current configuration to undo any unwanted changes you may have made.

Creating a Windows 7 Target

Although Windows 7 reached its end of life in January 2020, it still has a market share of approximately 2.25 percent, meaning there are potentially thousands of vulnerable Windows 7 computers in the world. Vulnerable computers are the favorite target of hackers, and an inherently vulnerable Windows 7 VM makes an excellent target for your penetration-testing lab.

You don't need to import an OVA to create the Windows 7 target. Instead, you'll build the VM in VirtualBox from scratch and use a Windows 7 .iso file (ISO) to perform the installation.

Acquiring a Windows 7 ISO File

Microsoft no longer provides a Windows 7 ISO file for download, so you will have to acquire the ISO file from another source. If you already have a Windows 7 ISO file, you may use it during the installation process. Otherwise, check your favorite Internet file repositories to find a Windows 7 ISO to download, or search online for available download locations such as https://tech-latest.com/download-windows-7-iso/. Download the Professional version of Windows if available; otherwise, download Ultimate or Home. Remember to scan all downloaded files with an antivirus program.

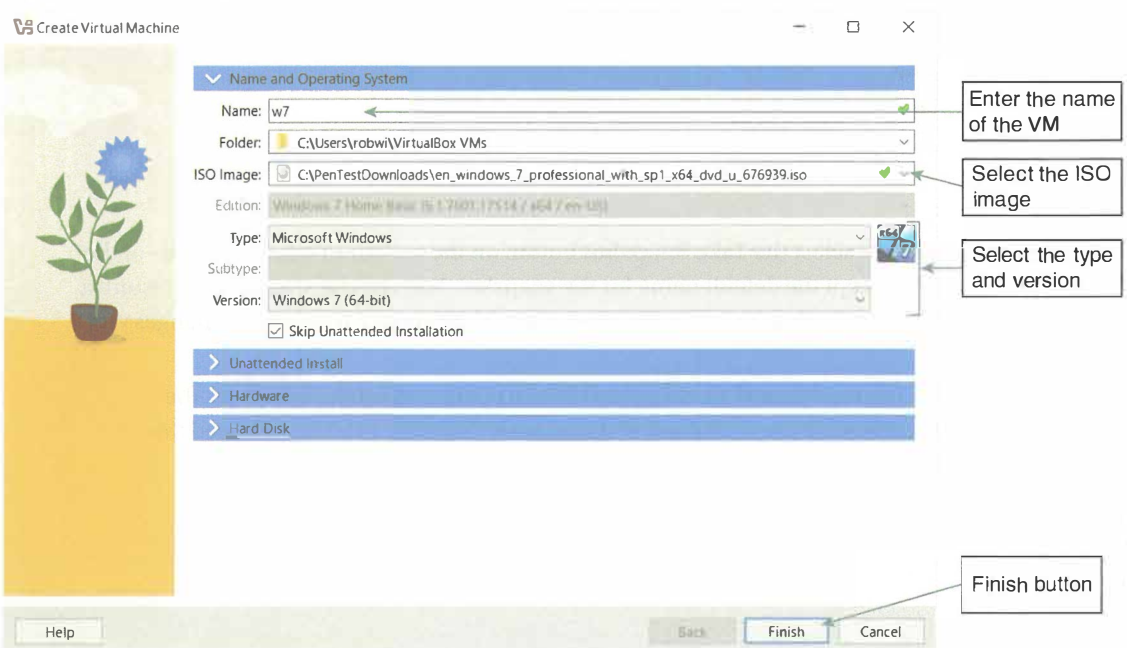
Creating a Windows 7 Target Virtual Machine

Creating a Windows 7 target VM involves two major steps. First, you create the Windows 7 VM, and then you install Windows 7. When you create the Windows 7 VM, you link the installation file to a virtual **DVD**, an optical media format for storing information. You use the virtual DVD to simulate installing Windows 7 from a bootable DVD.

To create the Windows 7 VM:

1. Click **Machine** on the menu bar in the Oracle VM VirtualBox Manager window, and then click **New**.
2. In the Create Virtual Machine dialog box, enter **w7** as the Name of your VM. In the ISO Image text box, navigate to and select your Windows 7 ISO file. Select **Microsoft Windows** as the Type, **Windows 7 (64-bit)** as the Version, and insert a checkmark in the Skip Unattended Installation check box (Figure 2-12).

Figure 2-12 Configuring the Windows 7 virtual machine



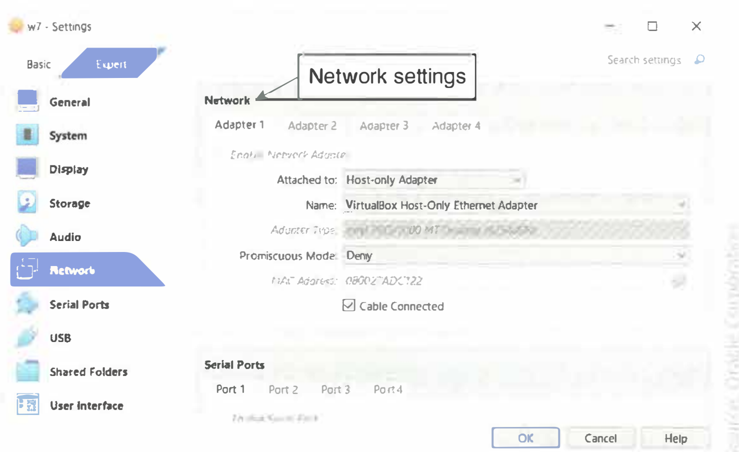
3. The default configurations in the Hardware and Hard Disk sections don't need to be changed, so click **Finish**.

To install Windows 7:

1. In the VirtualBox Manager window, click the **Start** button on the toolbar and then click **Normal Start** to start the w7 VM. Your VM boots from the virtual DVD and the Windows installation process begins.

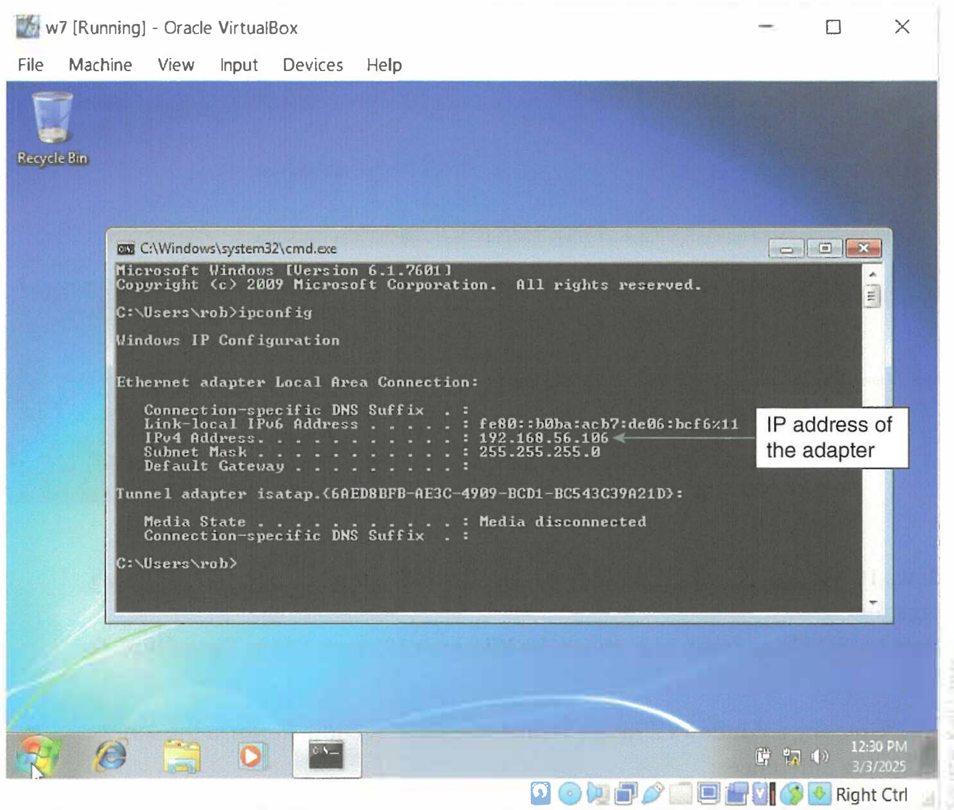
   If your attempt to start fails and you receive a USB-related message, click **USB** in the left pane of the w7 – Settings window, click the **Enable USB Controller** check box, and then click the **USB 1.1 (OHCI) Controller** option button to change the USB virtual hardware options.
2. When the Windows 7 installation begins, follow the instructions in the setup dialog boxes.
3. Choose to perform a custom installation and then choose the **Disk 0 Unallocated Space** as the location for installing Windows.
4. Make note of the username, password, and computer name you enter.
5. Select the **Skip** button in the Type your Windows product key window.
6. Select **Ask me later** in the Help protect your computer and improve Windows automatically window.
7. Choose **Home network** in the Select your computer's current location window.
8. In the VirtualBox Manager window, right-click the Windows 7 VM and select **Settings** on the short-cut menu. Click **Network** in the left pane, and then change the Network settings to attach the VM to the **Host-only Adapter** named **VirtualBox Host-Only Ethernet Adapter** (Figure 2-13). Click **OK**.

Figure 2-13 Adjusting network settings



9. Select the **w7** VM and then select the **Take Snapshot** button on the toolbar to take a snapshot of the VM. Provide a name or accept the default name for the snapshot. You can use snapshots to restore a VM to the state the VM was in when the snapshot was taken. This is useful to reset any changes you may make to the VM.
10. In the Windows 7 VM, use the `ipconfig` command from a command prompt, or check the adapter properties to determine the IP address of the adapter (Figure 2-14). You will need this IP address for future activities.

Figure 2-14 Using ipconfig to determine IP address details



Creating a Windows 10 Target

Even though Windows 10 is supposed to reach end of life in October 2025, it still has more market share (54 percent) than Windows 11. You are likely to work with many Windows 10 systems during real-world pen-testing activities.

You follow the same major steps to create a Windows 10 VM in VirtualBox as you do when creating the Windows 7 VM, except you use a Windows 10 ISO file to perform the installation.

To download a Windows 10 ISO file, you must first download the **Media Creation Tool**, a Microsoft utility used to download ISO files for operating systems.

To download a Windows 10 ISO file:

1. Use a web browser to go to **https://www.microsoft.com/en-ca/software-download/windows10**. Scroll down and select the **Download Now** button (Figure 2-15).
2. When the download is complete, open and run the **MediaCreationTool.exe** file. When the Applicable notices and license terms window opens, agree to the terms by clicking **Accept**.
3. When the What do you want to do? window is displayed, click the **Create installation media** option button and then click **Next**.
4. In the Select language, architecture, and edition window, accept the default settings by clicking **Next**.
5. In the Choose which media to use window, click the **ISO file** option button and then click **Next**.
6. In the Select a path dialog box, choose a download location, and then select the **Save** button. Note the default file name is Windows.iso. Remember the file name and location so you can find the ISO file later.

Figure 2-22 DVWA open in a web browser at main menu

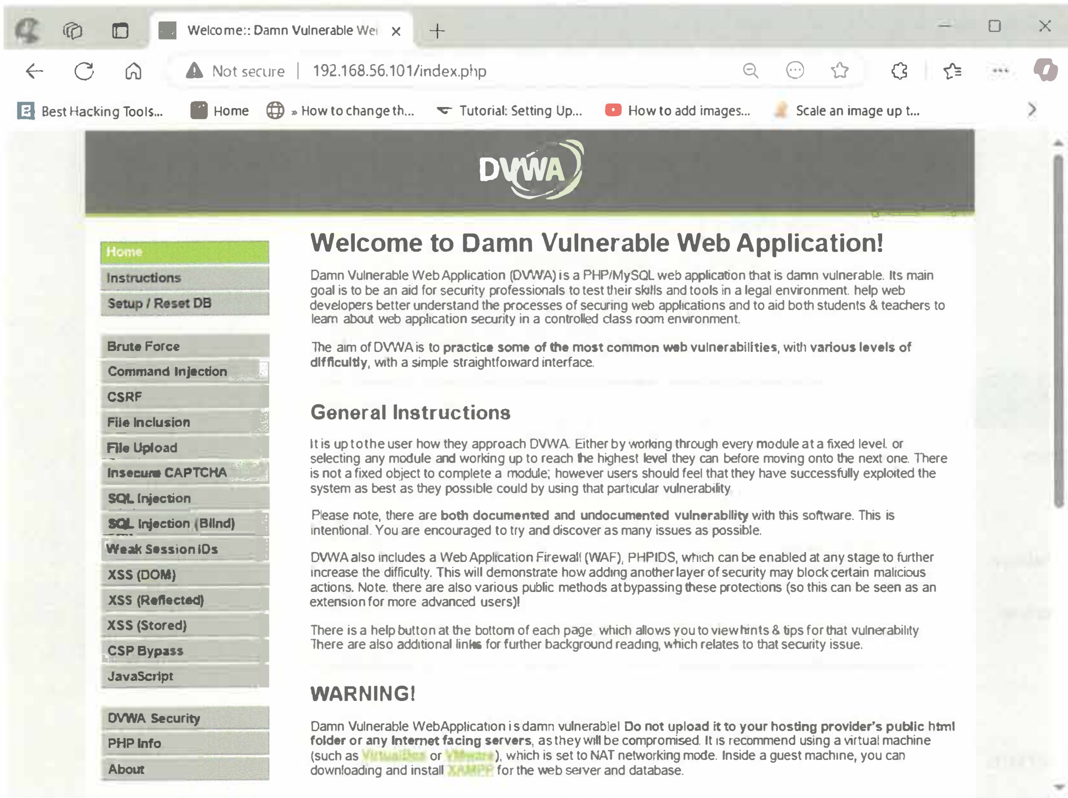
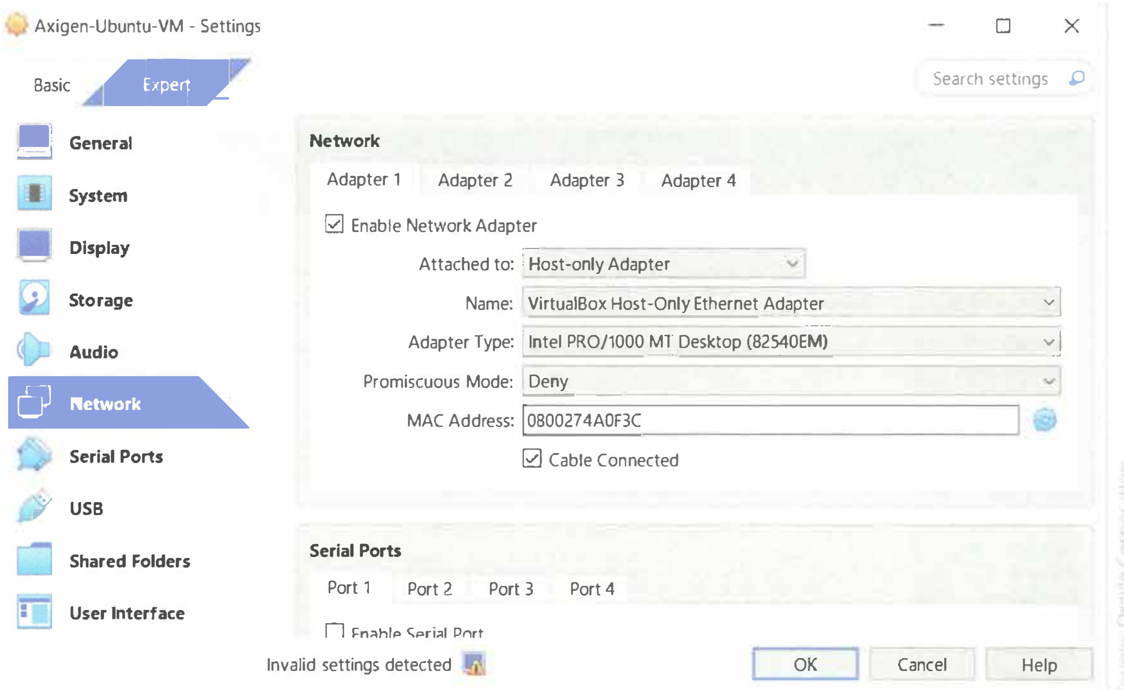


Figure 2-23 Downloading Axigen VMware image

3. In VirtualBox, click **File** on the menu bar, and then click **Import Appliance**. Navigate to the folder containing the OVA file you extracted, click the OVA file, and then click **Finish** to import Axigen into VirtualBox. Change the network settings so the Axigen VM is attached to the **Host-only Adapter** (Figure 2-24), and then click **OK**.

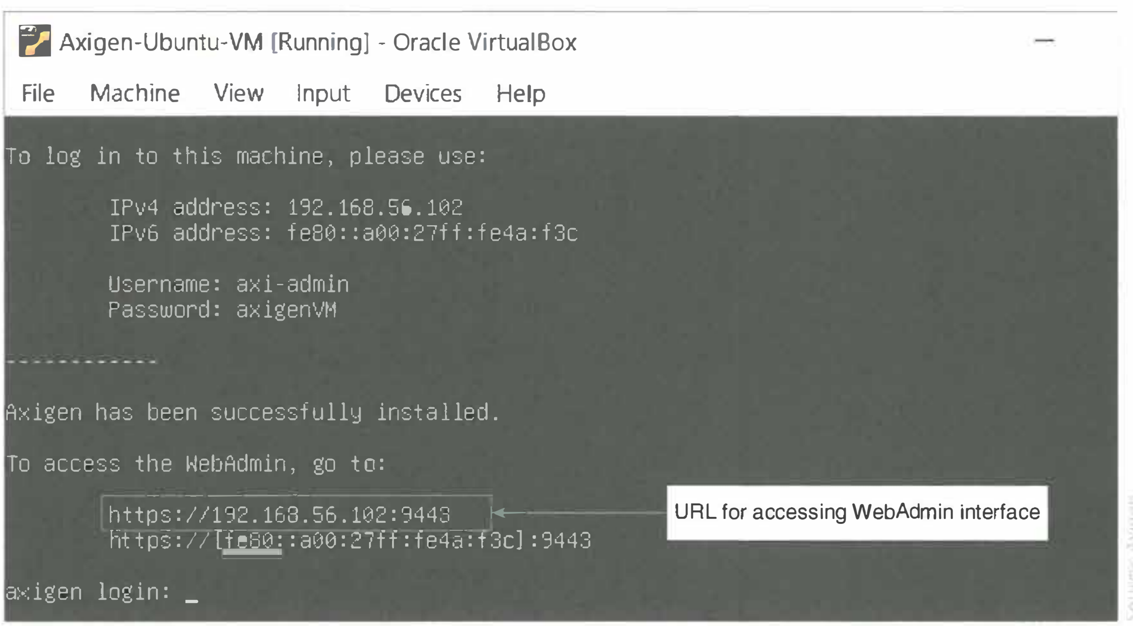
Figure 2-24 Setting Axigen Network to Host-only Adapter



To set up the mail server:

1. In the left pane of the VirtualBox Manager window, click the Axigen appliance, such as Axigen-Ubuntu-VM, click **Start** on the toolbar, and then click **Normal Start**.
2. After the Axigen appliance starts, a window resembling Figure 2-25 opens. Note the URL displayed for accessing the WebAdmin interface.

Figure 2-25 Axigen WebAdmin URL and IP address information



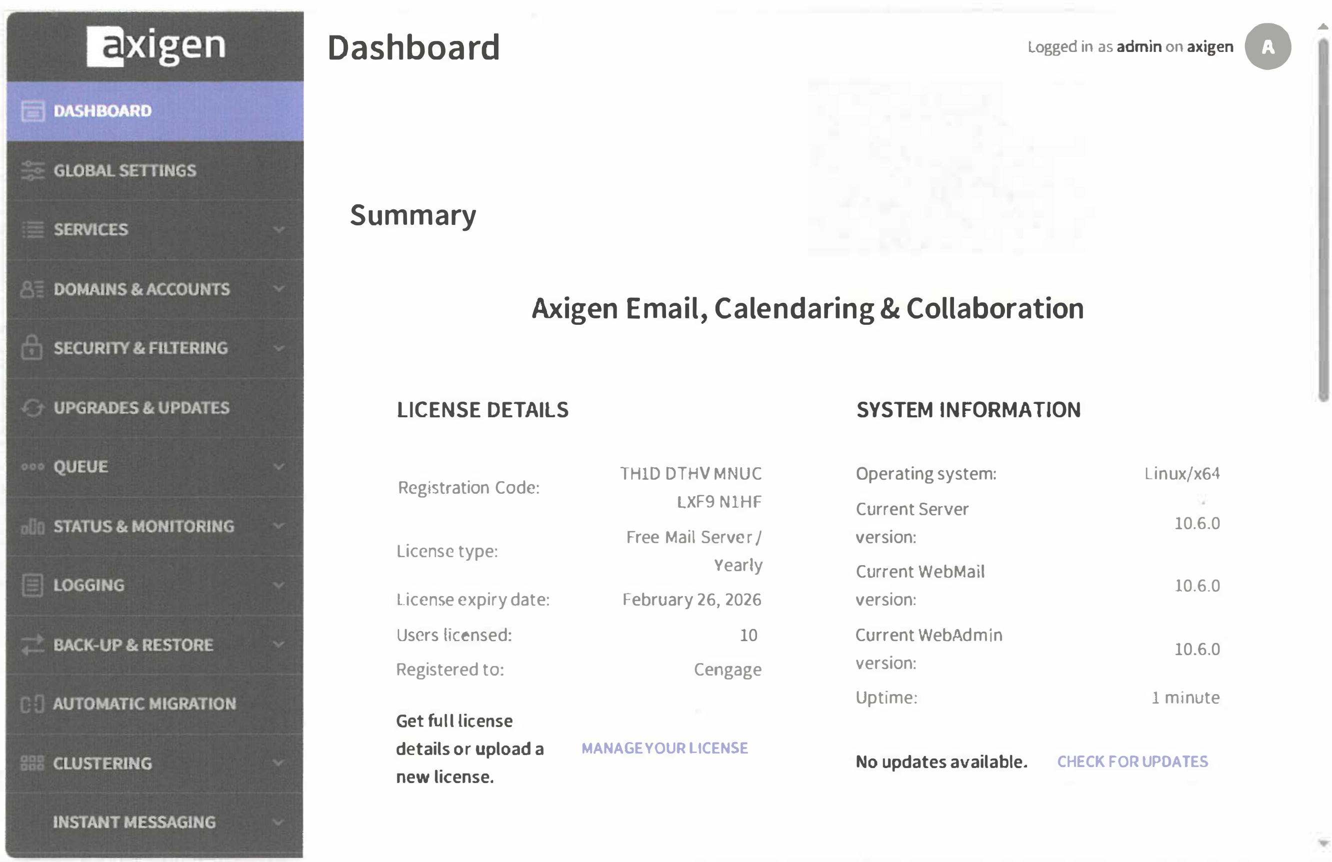
3. Follow the prompts to open your web browser, and then go to the WebAdmin URL you noted in Step 2. If you receive a warning that your connection is not private, click the **Advanced** button, and then click the **Proceed to** link.

4. Click the **I AGREE** button on the license agreement page, and then set your admin password. Use a secure and memorable password.
5. On the Your license page, choose **NEW FREE LICENSE**, and then fill in the Activate Your Free Mail Server form. Make sure you use a working email address as the license key will be sent to that email.

   If you do not receive a license, you can browse to https://www.axigen.com/mail-server/register/ and get one. Use the Skip this step option at the bottom of the license page because you now have a license and will apply it later.
6. On the Your Services page, click **CONTINUE**.
7. Enter **pentestlab.com** into the PRIMARY DOMAIN box to create an email domain. Choose a password and enter it into the POSTMASTER ACCOUNT PASSWORD box. Click **CONTINUE**.
8. Click **FINISH** on the You're all set page to display the WebAdmin Dashboard page (Figure 2-26). If you had to request a license key using the registration page, you can click the **APPLY LICENSE KEY** button and follow the instructions to apply the license key you received in an email message.

Note 5 | Axigen provides online resources for installing and performing an initial configuration. You can find these resources on the following webpages:
www.axigen.com/documentation/performing-the-initial-configuration-onboarding-p65437723
www.axigen.com/documentation/deploying-running-axigen-in-vmware-virtualbox-p58327042

Figure 2-26 Axigen WebAdmin dashboard

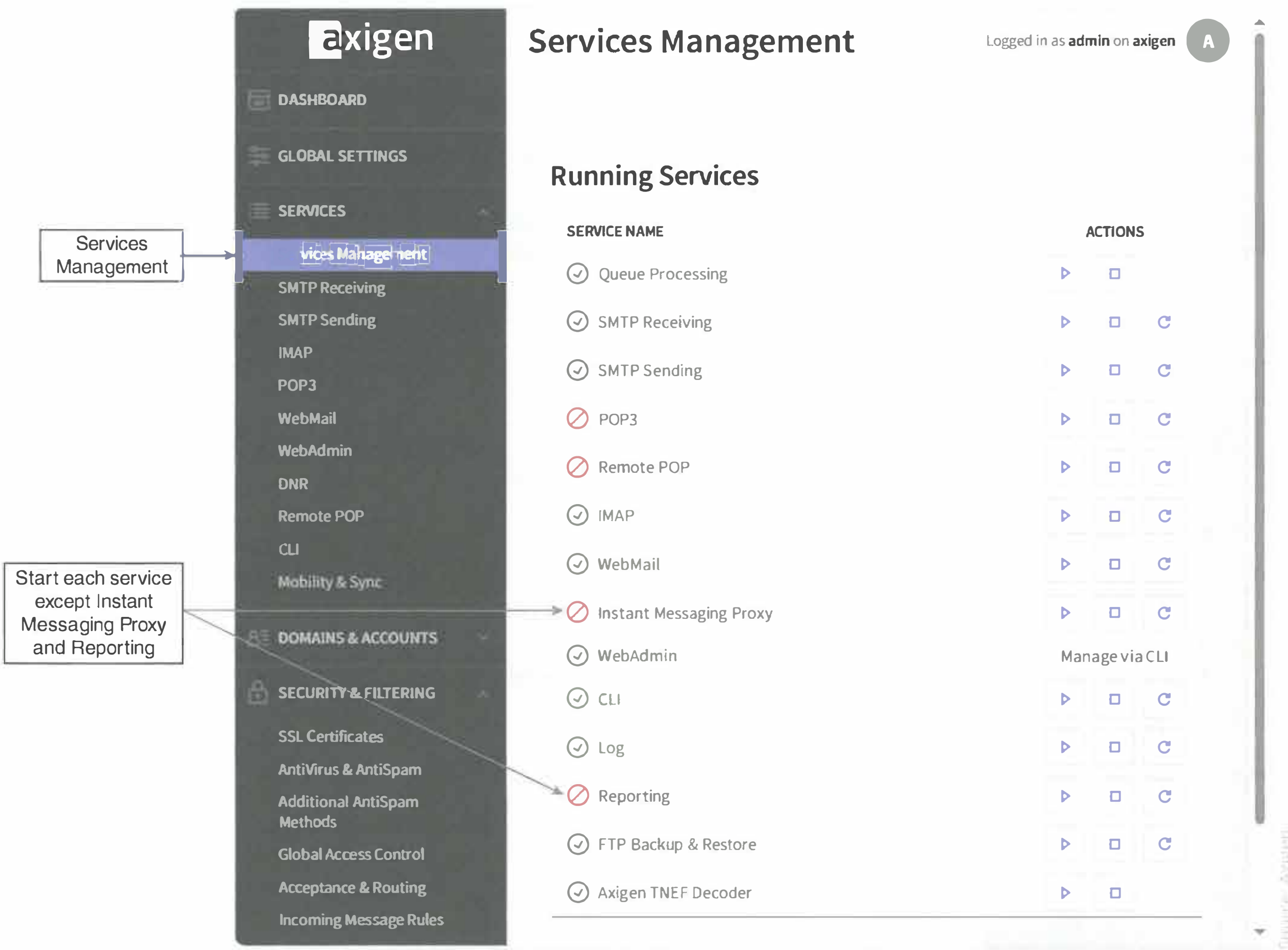


9. In the left pane of the Dashboard window, expand **SECURITY & FILTERING** and then click **Acceptance & Routing**. Scroll down and click to uncheck the **Activate Greylisting** check box to disable Greylisting. Click the **SAVE CONFIGURATION** button to save this change.

Caution! | If Greylisting is not disabled, the Axigen server will reject unauthenticated emails, and some activities in later chapters may not work. Greylisting is an anti-spam feature that rejects email messages from unknown senders. Enable all listeners; otherwise, some activities in later chapters may not work.

10. In the left pane of the Dashboard window, expand **SERVICES** and then click **Services Management** to display a list of running services. Click the arrow button to start each service except for Instant Messaging Proxy and Reporting (Figure 2-27). A green circle and checkmark indicate the service is running, and a red circle with slash indicates it is not. You do not need to click the arrow for services that are already running. In Figure 2-27, POP3 and Remote POP still need to be started.

Figure 2-27 Services Management window



11. In the left pane of the Services Management window, click **SMTP Receiving** to make sure SMTP receiving listeners are enabled.
12. In the left pane, expand **DOMAINS & ACCOUNTS** and then click **Manage Accounts** to add accounts to the Axigen server so that you have email addresses to use for future activities. Select the **pentestlab.com** domain in the DOMAIN list and then click the **ADD ACCOUNT** button. Complete the requested information, and then click **QUICK ADD**. You can create email addresses and accounts now, or you can return to this page later.
13. Use the VirtualBox Manager to check that the network adapter setting for this target VM is set to a **Host-only Adapter**. This is necessary so that this VM is on the same network as your other VMs.
14. Take a snapshot of this VM so that you can reset to this configuration to undo any unwanted changes you may have made.